

Viral Hepatitis – Summary

Key Symptoms: Jaundice, fever, arthralgia/myalgia, fatigue/malaise, upper right quadrant pain, dark urine, pale stools, etc.

Most common causing-pathogens: HAV, HBV, HCV, HDV, HEV

Diagnosis: LFTs, Bilirubin, Glucose, INR, serological/viral markers

Drug Causes: Acetaminophen, NSAIDs, Herbal Products, Statins, Antibiotics, TNF- α inhibitors, Interferon α /beta, etc.

Non-Pharmacological Treatment:

- **Acute Hepatitis (< 6 months):** Avoid alcohol for at least 3 months or until liver enzymes and hepatic function are completely normalized
- **Chronic Hepatitis (> 6 months):** Avoid/minimize alcohol consumption, weight loss (if BMI > 30), optimize blood sugar in diabetes and encourage smoking cessation
- **Prevention:** vaccination for HAV/HBV (currently no vaccine for HCV), HBV vaccination confers immunity against HDV

Pharmacological Treatment:

- **Acute Hepatitis**
 - In HBV, consider antiviral therapy with Nucleos(t)ide analogues
 - If INR > 1.4, give Vitamin K 10mg PO/SC
 - If experiencing encephalopathy, give lactulose targeting to 2-3 loose bowel movements per day
- **Chronic HBV**
 - Treat those who are most likely to progress to hepatocellular carcinoma (HCC) and/or cirrhosis, those with HIV co-infection, those with HBV related complications (cirrhosis, renal failure)
 - Peginterferon- α -2a SC weekly x 48 weeks **OR** nucleos(t)ide analogues (e.g. tenofovir DF/AF, entecavir, etc.) daily
- **Chronic HCV**
 - Sofosbuvir/Velpatasvir **OR** Sofosbuvir/Ledipasvir **OR** Glecaprevir/Pibrentasvir **OR** Sofosbuvir/Velpatasvir/Voxilaprevir +/- Ribavirin
 - Regimen depends on the genotype, hx of treatment and hx of cirrhosis

Monitoring Parameters:

- **Chronic Hepatitis:** Achieve Sustained Virologic Response (no HBV DNA/HCV RNA), prevent cirrhosis and HCC, and achieve HBeAg/HBsAg seroconversion (HBV only)

Additional Key Points:

- **HBV in pregnancy:** if high viral load, start TDF or TAF asap, no later than 2nd trimester. Monitor LFTs closely (ALT q 4 weeks for the first 3 months, then at 6 months, then routinely afterwards).
- Interferons have the worst side effect profile (e.g., flu-like symptoms, hair loss, memory loss, neutropenia, diarrhea, weight loss, infections, etc.) and multiple contraindications (e.g., pregnancy, decompensated liver disease, HIV, severe cardiac disease, solid organ transplant (except liver))
- When patients begin immunosuppressive therapy, Hep B viral load rises and may result in flare-ups
- Co-infection in patients with HIV has negative outcomes, treat as soon as possible. Do not use nucleoside analogues as monotherapy due to risk of resistant HIV strains. Watch for interactions.
- Ribavirin should be used with caution as it can cause hemolytic anemia (dose-dependent)